Audio Electronic Device Measurement Report

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Shannon, Sihyeon, Lee, Ta'mya, Erin

Disclaimer: Because of a scheduling conflict we worked together with another group (Lee, Ta'mya, and Erin) to perform all the measurements.

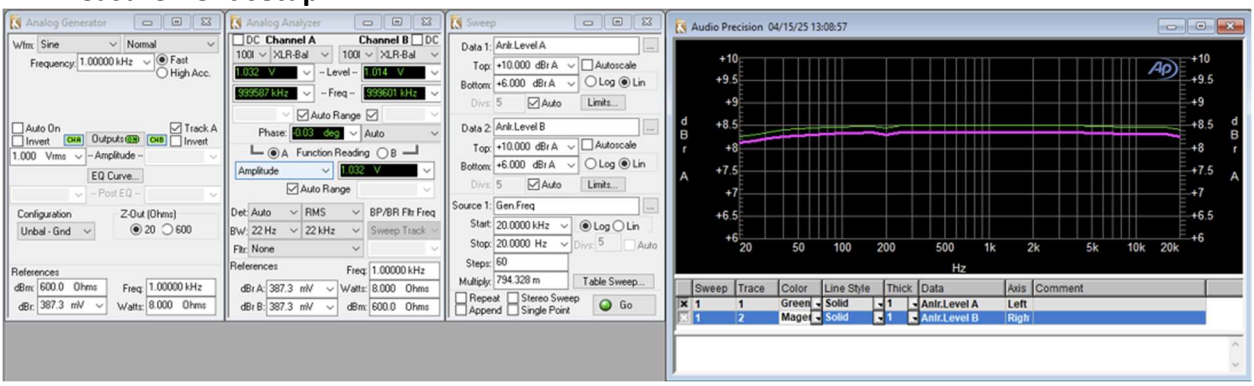
Device Under Test 1: Focusrite Scarlett 2i4

1. Manufacturer's Specifications

- Frequency response: 20 Hz – 20 kHz ±0.1 dB
- THD+N: <0.01% ('A'-weighted, 16 dB input gain, -1 dBFS output,
- 20 Hz – 22 kHz filter)
- Phase: Not specified
- Crosstalk: <80 dB
- Signal-to-Noise ratio: 105 dB, 'A'-weighted

Reference: https://fael-downloads-prod.focusrite.com/customer/prod/s3fs-public/focusrite/downloads/8174/scarlett2i4-user-guide.pdf

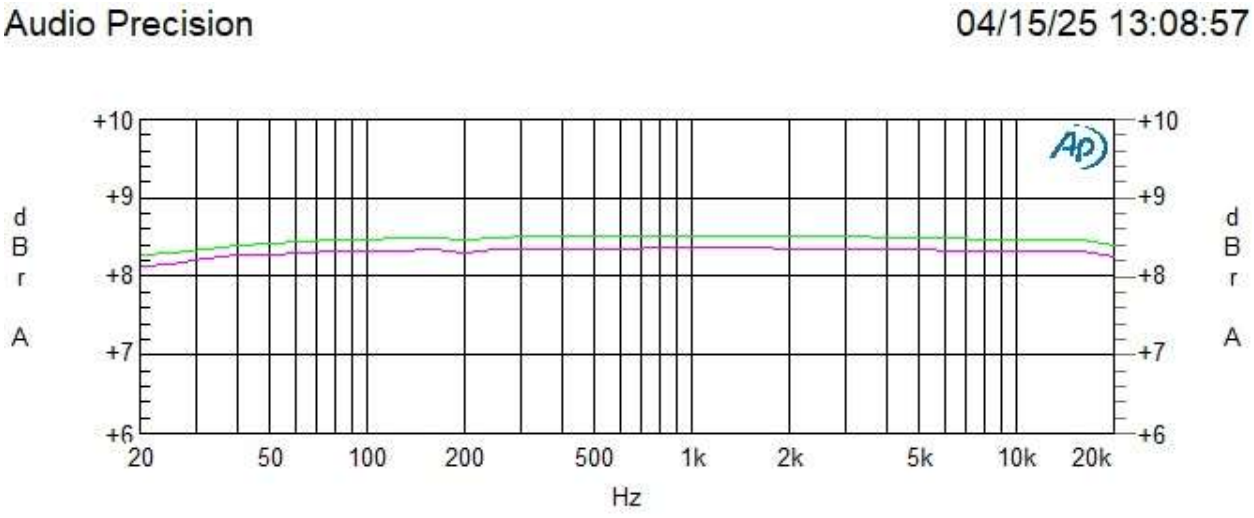
2. Measurement setup



After setting up the cables for inputs and outputs, we turned the device on. To set generator and analyzer references, we changed "configuration" on the analog generator to "Unbal – Gnd". Outputs were turned on for both channels. For the analyzer, we turned the pre-amp up until we reached close to 1V on level for both channels. Also in analyzer, changed bandwidth (BW) to 22Hz – 22kHz. Then went to edit tab in ribbon to "Set Analyzer dBr Ref" and "Set generator dBr Ref". For sweep panel, data 1 was set to "Anlr.LevelA", and data 2 was set to "Anlr.LevelB". Changed top and bottom for both data 1 and 2 to be dBrA units. Ensured start and stop frequency were at 20kHz and 20Hz respectively. Changed steps to 60 then clicked on "Go".

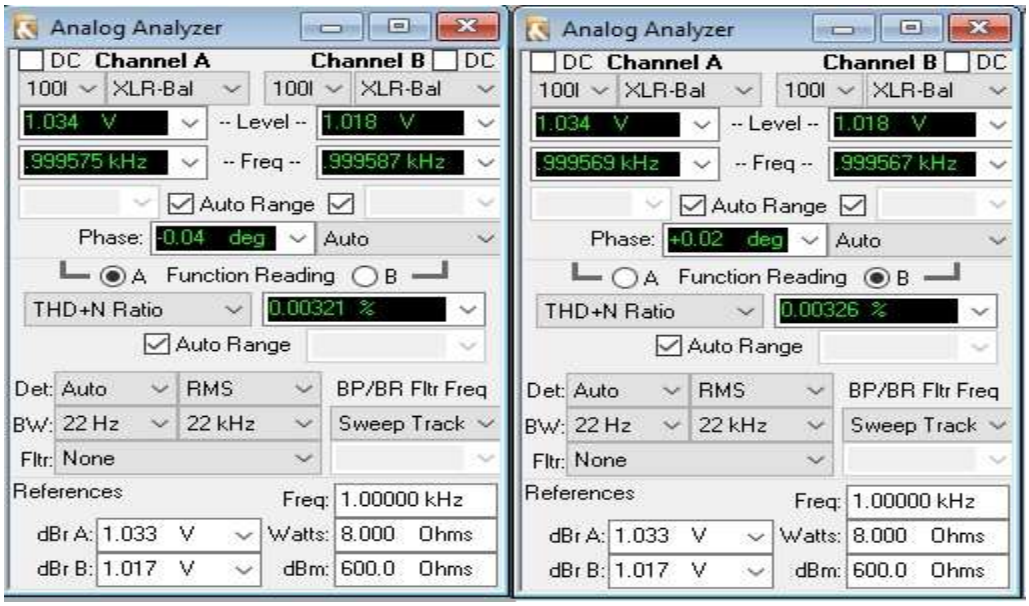
3. Results

3.1. Frequency response of Ch 1 & Ch 2 showed a very good flat line. We could see a slight difference in level of around 0.1 dB between the two channels. This level difference is very minute and not significant as a level difference of 0.1 dB is very difficult for human ears to detect, so the device is providing a good frequency response.

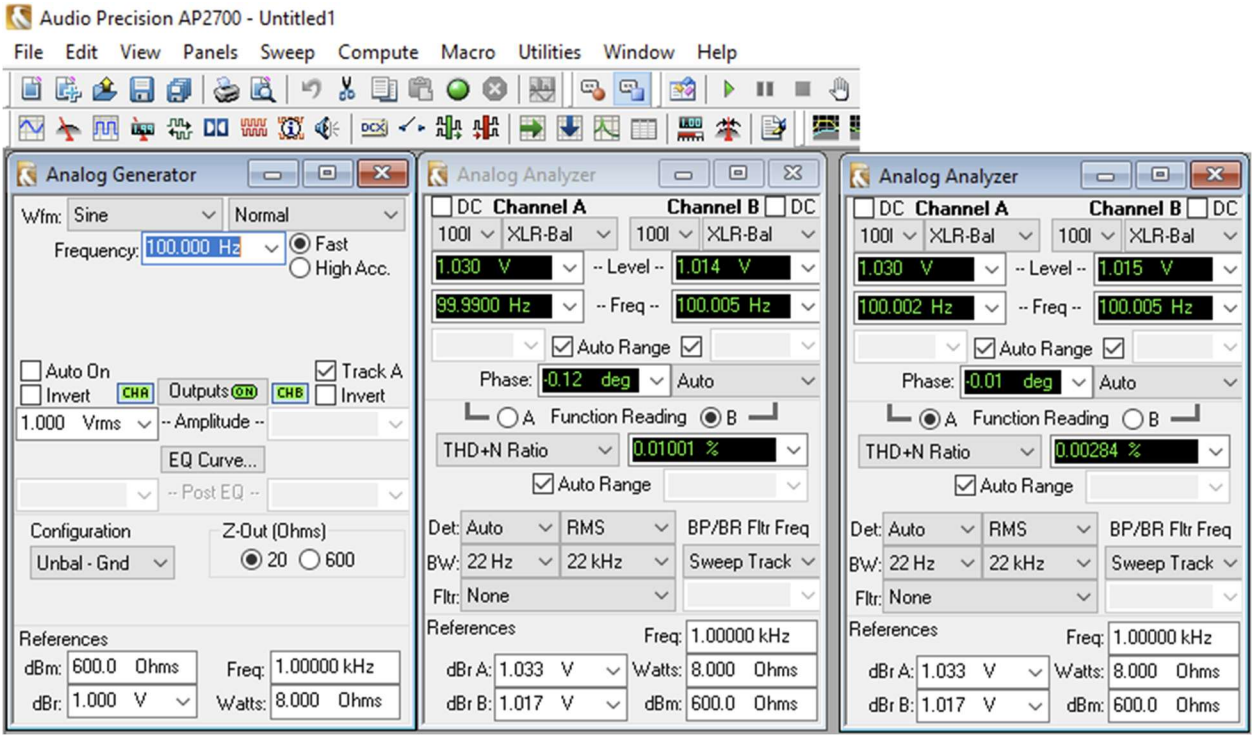


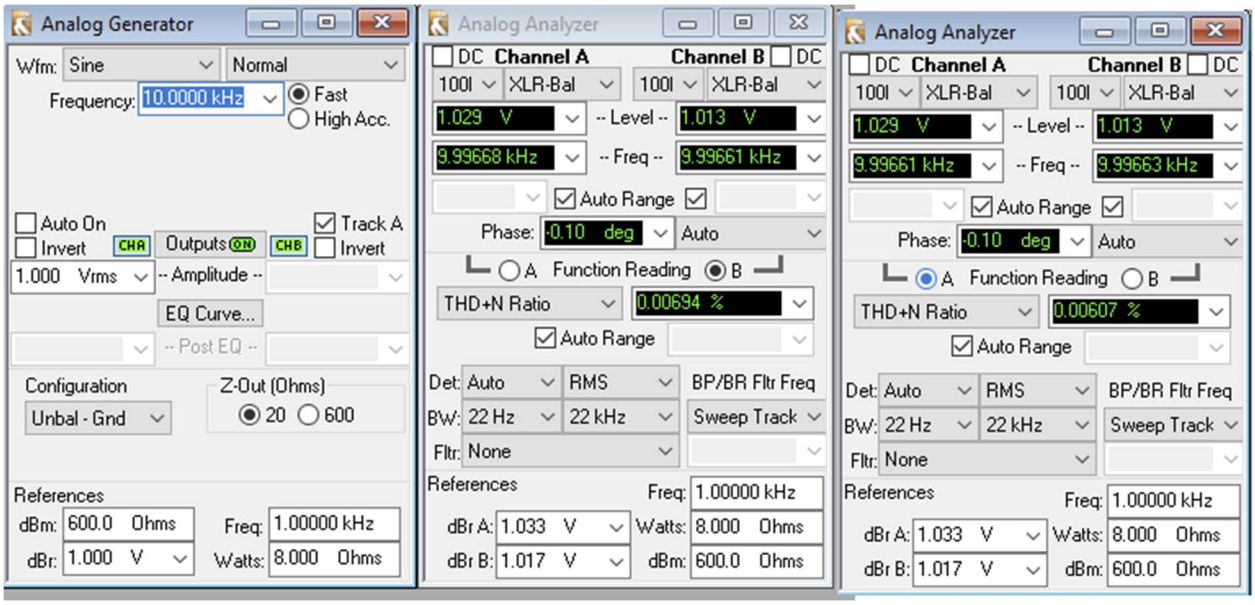
Sweep	Trace	Color	Line Style	Thick	Data	Axis	Comment
1	1	Green	Solid	1	Anlr.Level A	Left	
1	2	Magenta	Solid	1	Anlr.Level B	Right	

3.2. THD+N of channel A is 0.00321%, and channel B is 0.00326%.



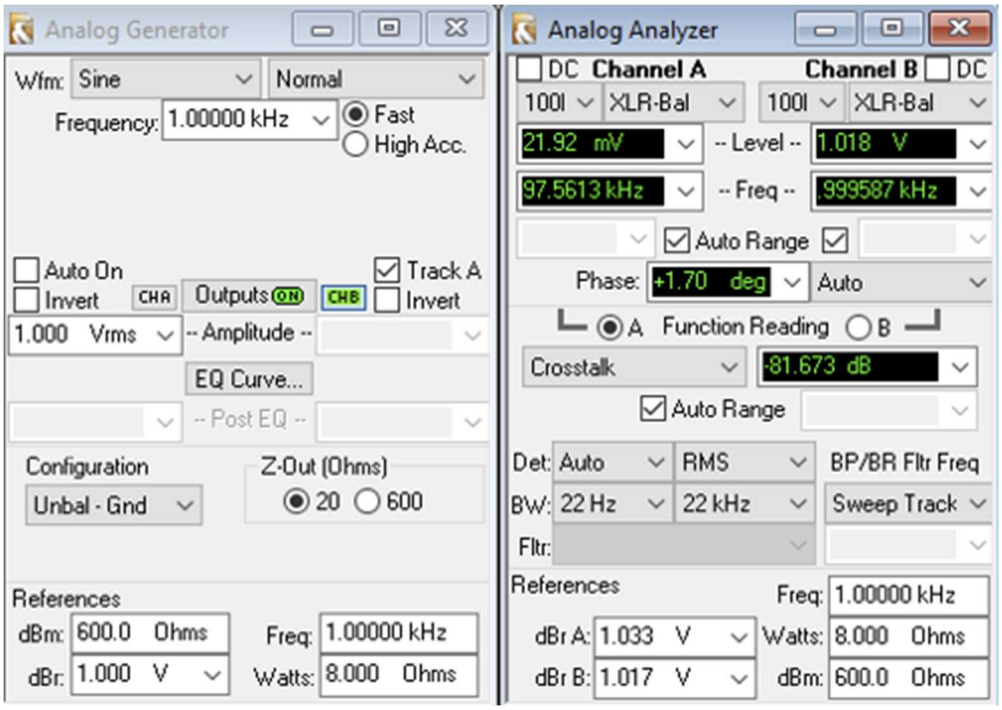
3.3. When reading channel A, the phase difference between the two channels at 100Hz is -0.12 deg. And when reading channel B, the phase difference between the two channels is -0.01 deg. Both these values are very tiny, indicating that the signals are in phase. When reading channel A, the phase difference between the two channels at 1kHz is -0.02 deg. And when reading channel B, the phase difference between the two channels is -0.07 deg. Again, both these values are extremely tiny, which indicates that the signals are in phase. For 10kHz, both channel A and channel B readings showed the phase difference between the two channels is -0.10 deg. Again, these values are very tiny, so even at higher frequencies the two channels remain in phase.

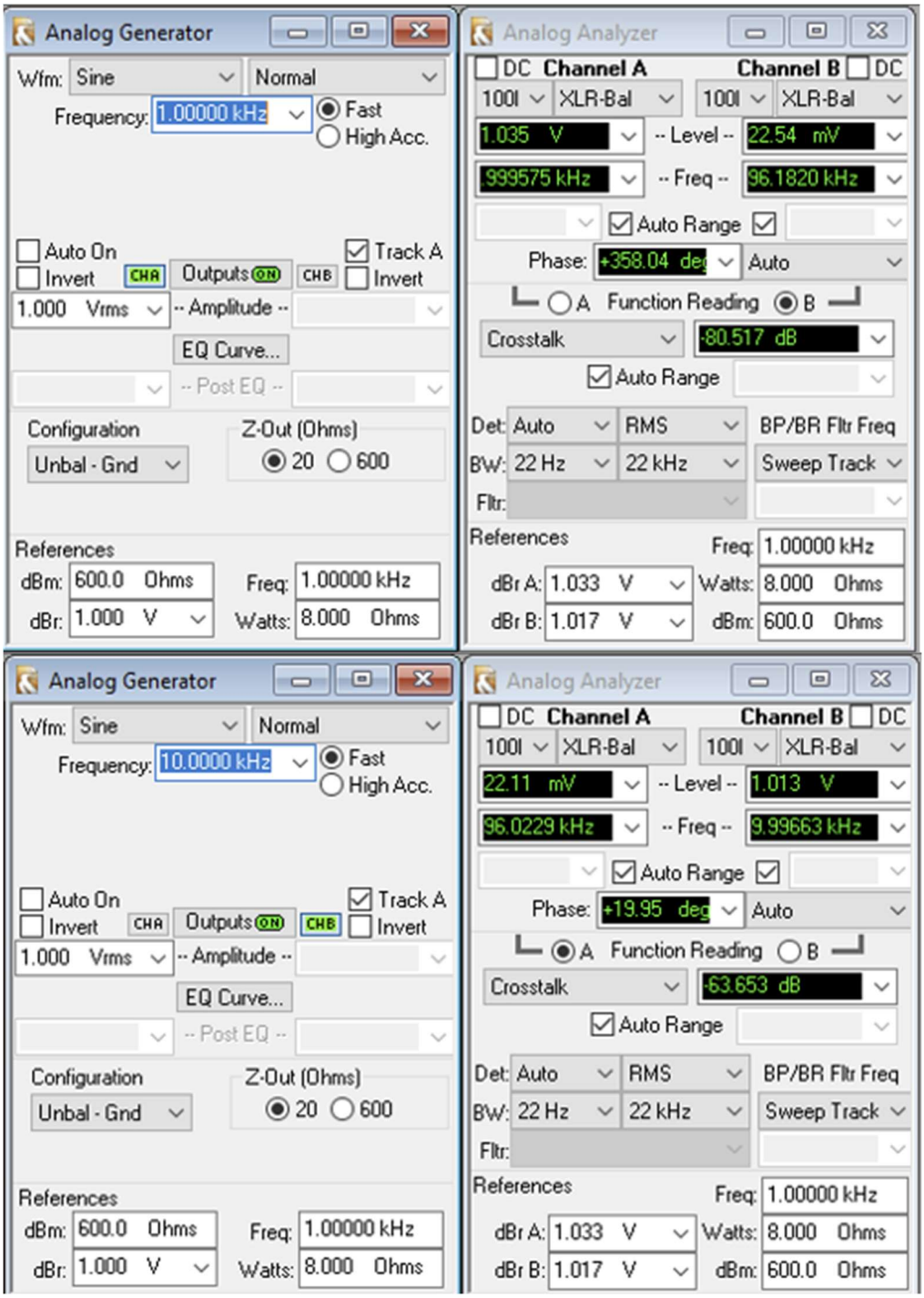


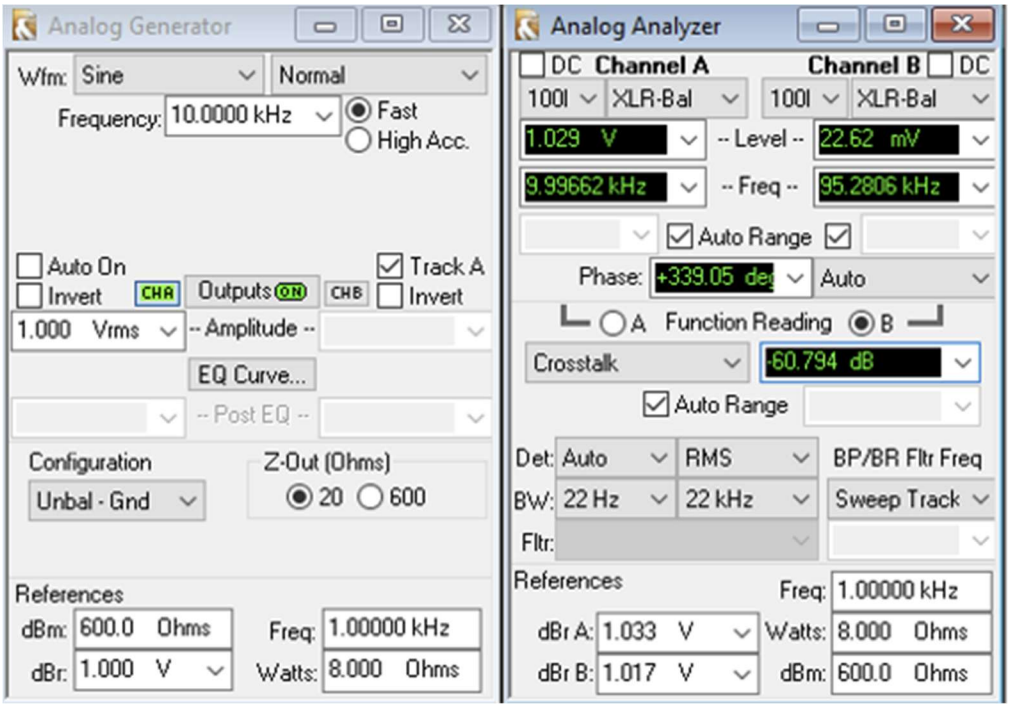


3.4. For 1kHz, crosstalk from B into A is -81.673 dB, and crosstalk from A into B is -80.517 dB. For 10kHz, crosstalk from B into A is -63.653 dB, and crosstalk from A into B is -60.794 dB. Crosstalk is greater at the higher frequency, but all values are less than -50 dB which indicates that channel separation is very good.

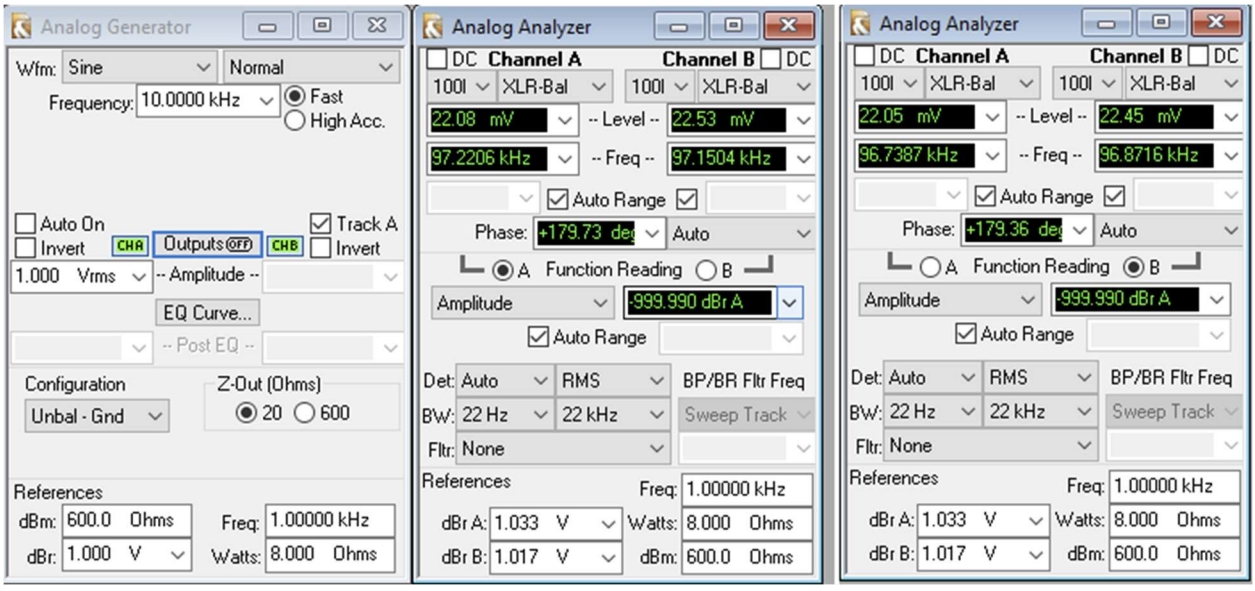
Reference: https://www.edn.com/crosstalk-how-much-is-too-much-rule-of-thumb-19/#:~:text=Spoiler%20summary%3A%20In%20single%2Dended,%2D50dB%2C%20or%200.3%25.




Analog Generator
Wfm: Sine
Normal
Frequency: 1.00000 kHz
Fast
High Acc.
Auto On
Invert
CHA
Outputs ON
CHB
Track A
Invert
1.000 Vrms
-- Amplitude --
EQ Curve...
-- Post EQ --
Configuration
Unbal - Gnd
Z-Out (Ohms)
20
600
References
dBm: 600.0 Ohms
dBr: 1.000 V
Freq: 1.00000 kHz
Watts: 8.000 Ohms
Analog Analyzer
DC Channel A
Channel B
DC
100I
XLR-Bal
1.035 V
-- Level --
22.54 mV
999575 kHz
-- Freq --
96.1820 kHz
Auto Range
Phase: +358.04 deg
Auto
A
Function Reading
B
Crosstalk
-80.517 dB
Auto Range
Det: Auto
RMS
BP/BR Fltr Freq
BW: 22 Hz
22 kHz
Sweep Track
Fltr:
References
Freq: 1.00000 kHz
dBr A: 1.033 V
Watts: 8.000 Ohms
dBr B: 1.017 V
dBm: 600.0 Ohms
Analog Generator
Wfm: Sine
Normal
Frequency: 10.0000 kHz
Fast
High Acc.
Auto On
Invert
CHA
Outputs ON
CHB
Track A
Invert
1.000 Vrms
-- Amplitude --
EQ Curve...
-- Post EQ --
Configuration
Unbal - Gnd
Z-Out (Ohms)
20
600
References
dBm: 600.0 Ohms
dBr: 1.000 V
Freq: 1.00000 kHz
Watts: 8.000 Ohms
Analog Analyzer
DC Channel A
Channel B
DC
100I
XLR-Bal
22.11 mV
-- Level --
1.013 V
96.0229 kHz
-- Freq --
9.99663 kHz
Auto Range
Phase: +19.95 deg
Auto
A
Function Reading
B
Crosstalk
-63.653 dB
Auto Range
Det: Auto
RMS
BP/BR Fltr Freq
BW: 22 Hz
22 kHz
Sweep Track
Fltr:
References
Freq: 1.00000 kHz
dBr A: 1.033 V
Watts: 8.000 Ohms
dBr B: 1.017 V
dBm: 600.0 Ohms



3.5. The signal-to-Noise ratio of both channels A and B is -999.990 dBr A, which indicates that the system was likely unable to calibrate. This could have been because the noise of the device is very low. This indicates an excellent signal-to-noise ratio.



4. Comparison between measurement results and the manufacturers' specifications.

	Specified on manual	Measured value
Frequency response	20 Hz – 20 kHz ±0.1 dB	Flat with 0.1 dB level difference
THD+N	<0.01%	Channel A: 0.00321% Channel B: 0.00326%
Phase at 100Hz	Not Specified	Channel A: -0.12 deg Channel B: -0.01 deg
Phase at 1kHz	Not Specified	Channel A: -0.02 deg Channel B: -0.07 deg
Phase at 10kHz	Not Specified	Channel A: -0.10 deg Channel B: -0.10 deg
Crosstalk at 1kHz	<80 dB	Channel A to B: -80.517 dB Channel B to A: -81.673 dB
Crosstalk at 10kHz	<80 dB	Channel A to B: -60.794 dB Channel B to A: -63.653 dB
SNR	105 dB, 'A'-weighted	Channel A: -999.990 dBr A Channel B: -999.990 dBr A

For the Focusrite Scarlett 2i4, the values specified on the user manual closely aligns with the values we measured. The measured frequency response of 0.1 dB is within the uncertainty limit specified. Measured values for THD+N for both channels A and B are below the specification of 0.01%. There were no specifications for phase, but all the phase values at 100Hz, 1kHz, and 10kHz were less than 1 degree which suggests that both channels are in phase. The specified crosstalk was <80dB and we can see that at 1kHz the measured numbers are very close to this value. However, at 10kHz crosstalk is greater than the specification. Since the manual does not indicate a specific frequency for crosstalk, it is likely that the manufacturer's stated value applies to lower frequencies. The specified SNR was 105db, but the software failed to calibrate it, so we can guess that the SNR is around 110 dB as the device noise is very low.

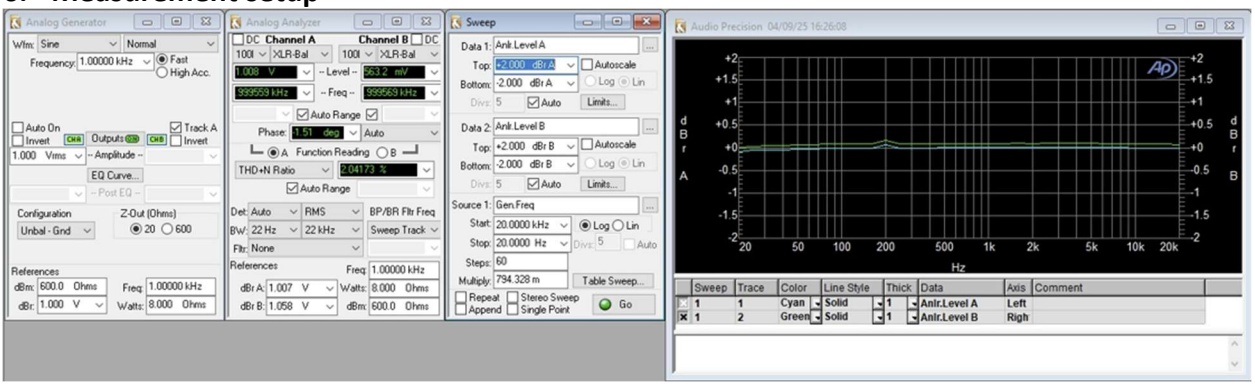
Device Under Test 2: Focusrite Scarlett 2i2

5. Manufacturer's Specifications

- Frequency response: 20 Hz to 20 kHz, ±0.1 dB
- THD+N: <0.003% (minimum gain, -1dBFS input with 22 Hz/22 kHz bandpass filter)
- Phase: Not specified
- Crosstalk: Not specified
- Signal-to-Noise ratio: Not specified

Reference: https://fael-downloads-prod.focusrite.com/customer/prod/downloads/Scarlett%202i2%202nd%20Gen%20User%20Guide%20v1.1%20English%20-%20EN.pdf

6. Measurement setup

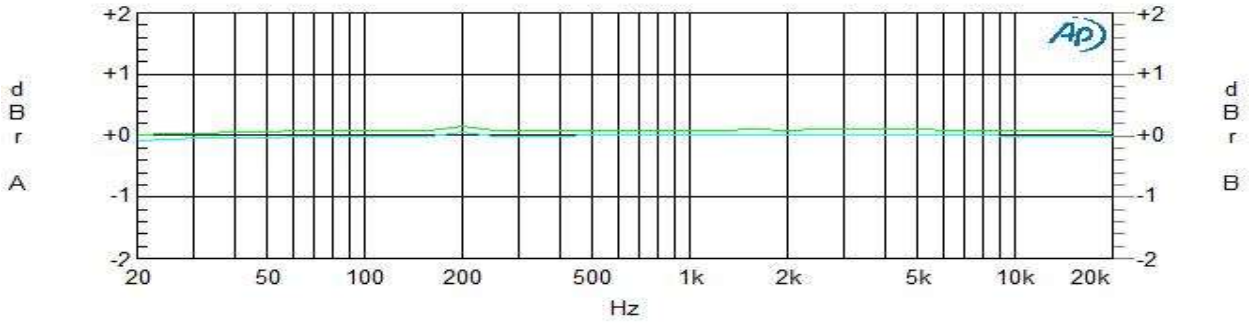


We switched the cables to work with Scarlett 2i2 and restarted the software. Like before, we changed "configuration" on the analog generator to "Unbal – Gnd". Outputs were turned on for both channels. For the analyzer, we turned the pre-amp up until we reached close to 1V on level. For this device, we were not able to align channel A and B to the same level. So, we only aligned channel A close to 1V. Changed bandwidth (BW) to 22Hz – 22kHz. Hit F3 and F4 on keyboard to "Set Analyzer dBr Ref" and "Set generator dBr Ref". For sweep panel, data 1 was set to "Anlr.LevelA", and data 2 was set to "Anlr.LevelB". Changed top and bottom for both data 1 and 2 to be dBr units. Changed steps to 60 then clicked on "Go".

7. Results

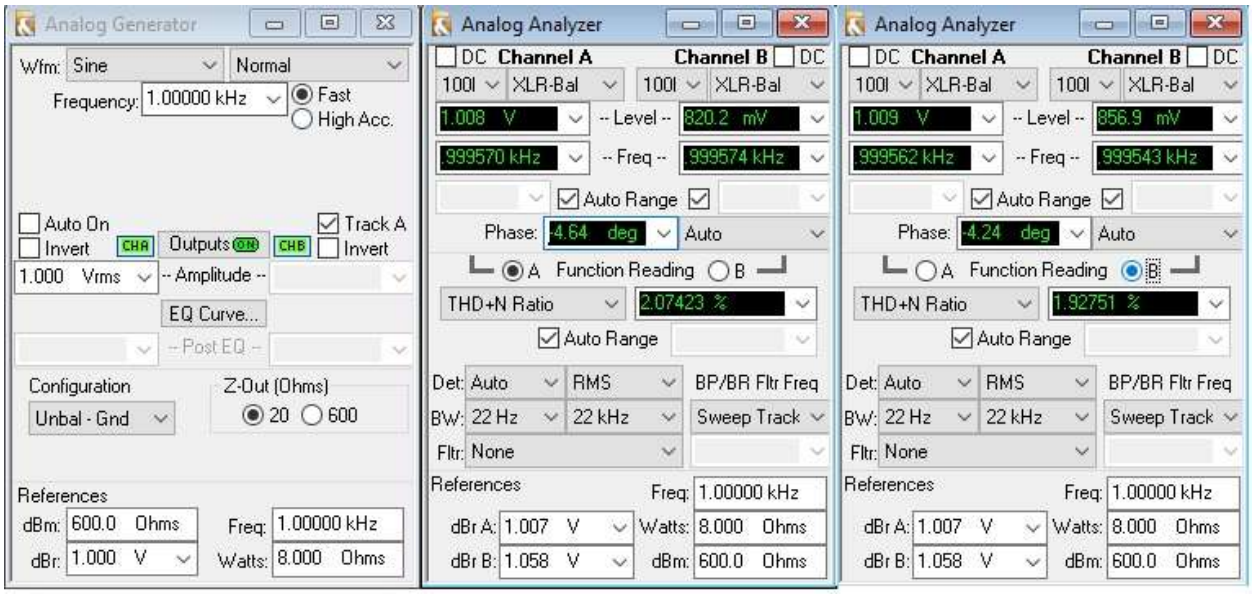
7.1. Frequency response curve shows a 0.1 dB boost at 200 Hz and some unevenness from 1k to 5k for both channels A and B. A very slight difference in level can be seen between the two channels.

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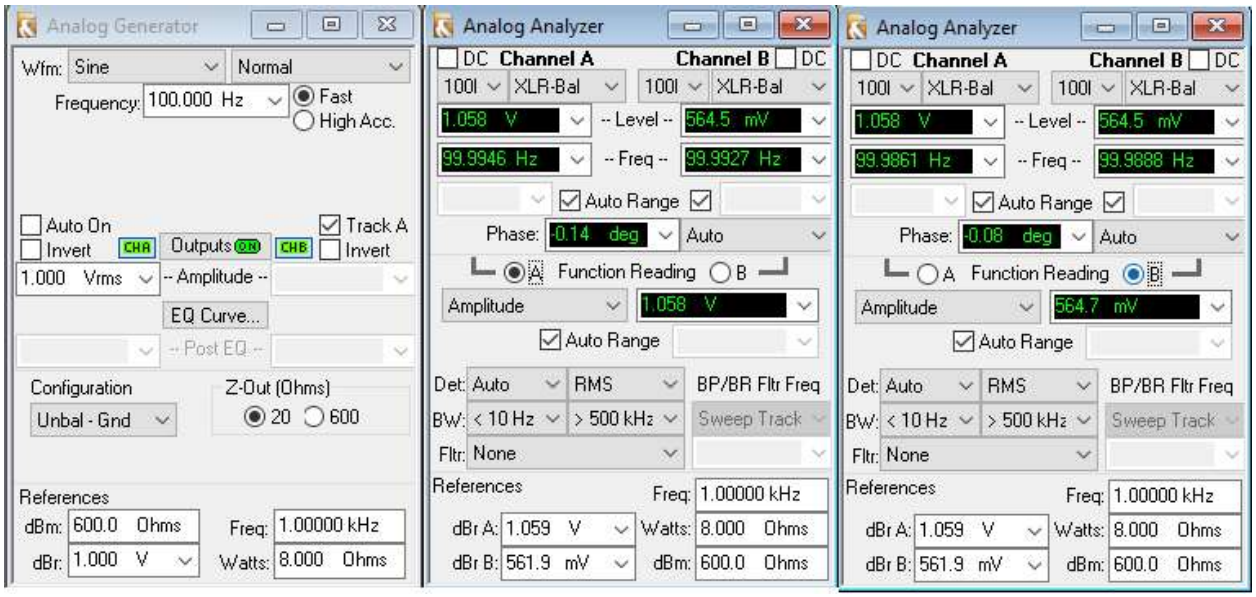


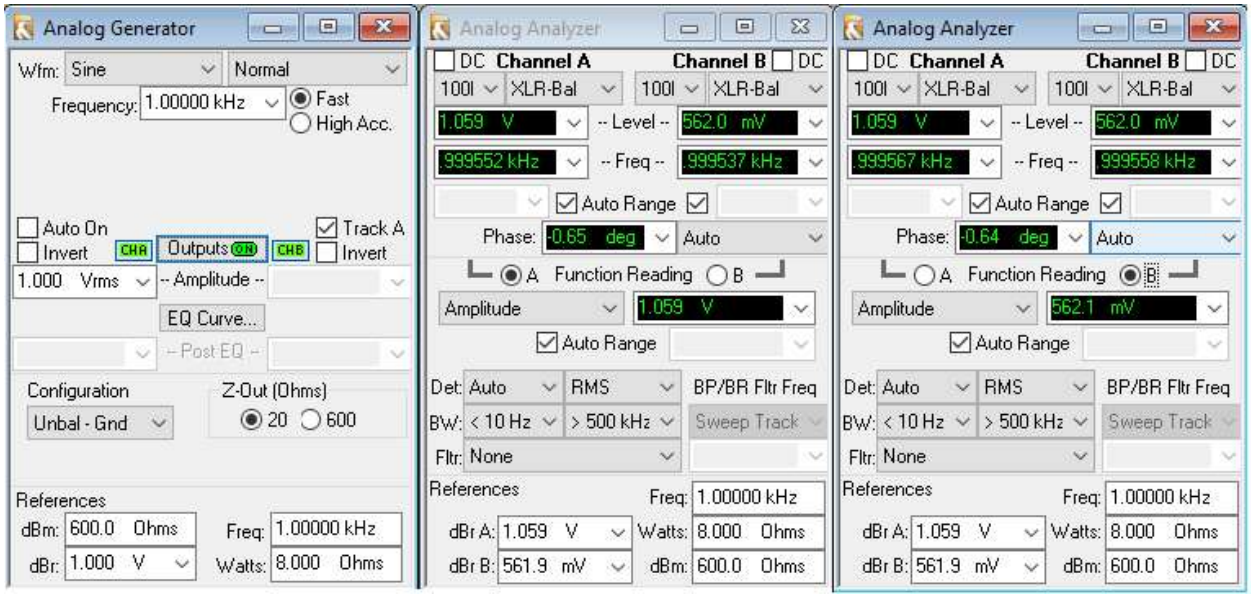
Sweep	Trace	Color	Line Style	Thick	Data	Axis	Comment
1	1	Cyan	Solid	1	Anlr.Level A	Left	
1	2	Green	Solid	1	Anlr.Level B	Right	

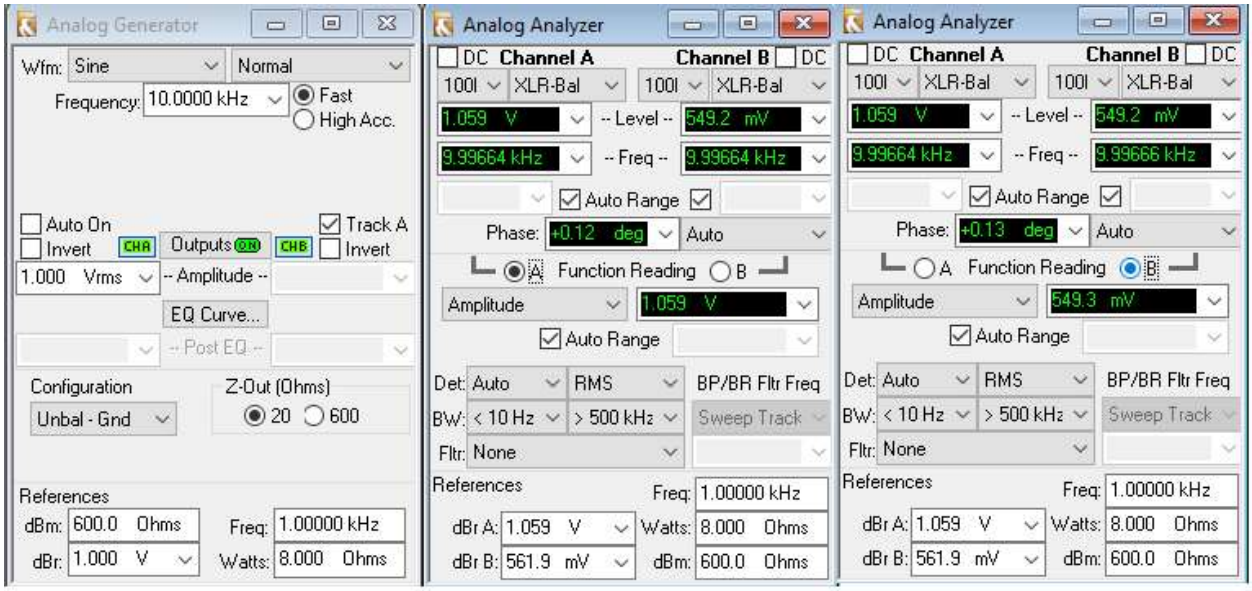
7.2. THD+N of channel A is 2.07423%, and channel B is 1.92751%. THD+N for both channels are higher than that specified on the manual suggesting that the device may be faulty.



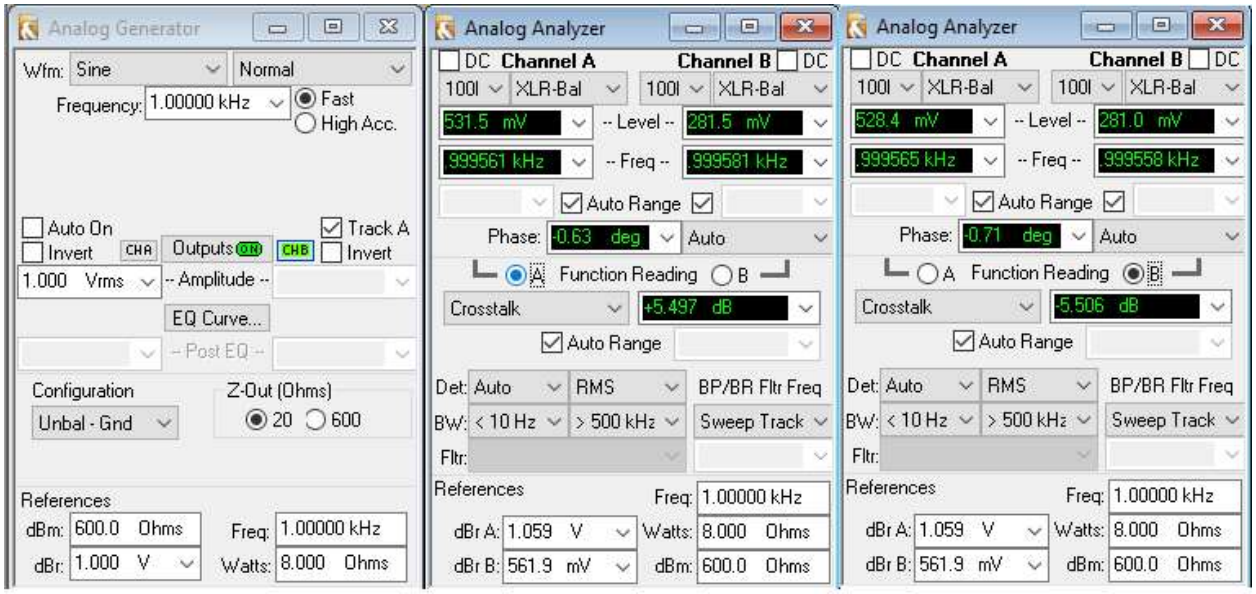
7.3. For phase and subsequent measurements, we overlooked one of the settings and forgot to change the BW because we had to restart the software multiple times to fix level fluctuations. The phase reading from channel A at 100Hz is -0.14 deg. And the phase reading from channel B is -0.08 deg. For 1kHz, phase reading of channel A is -0.65 deg and channel B is -0.64 deg. And for 10kHz, phase reading of channel A is +0.12 deg and channel B is +0.13 deg. Despite having significant THD+N and the levels not matching on both channels, we can see that the two channels are still in phase and the degree of separation is less than 1 degree.

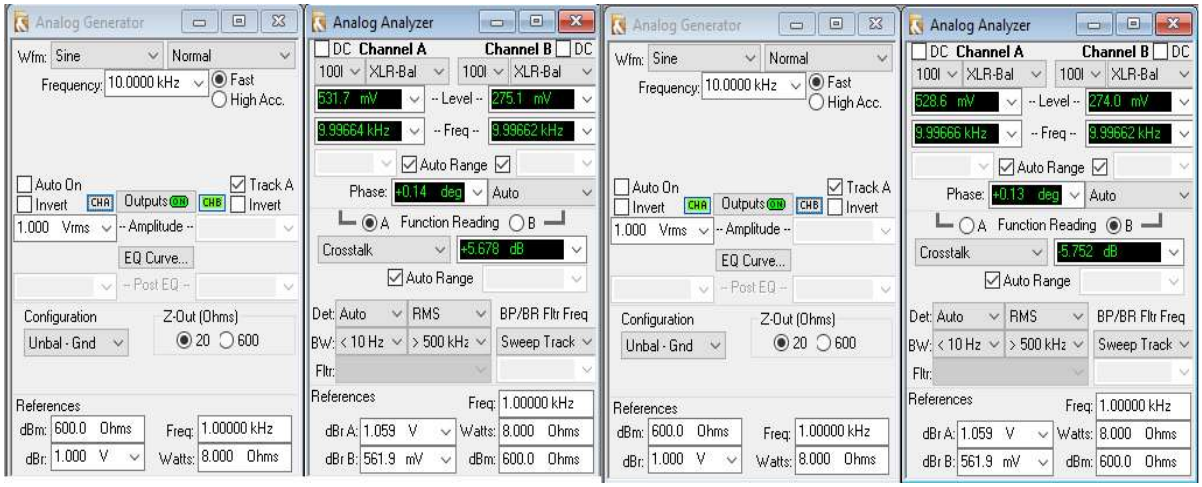




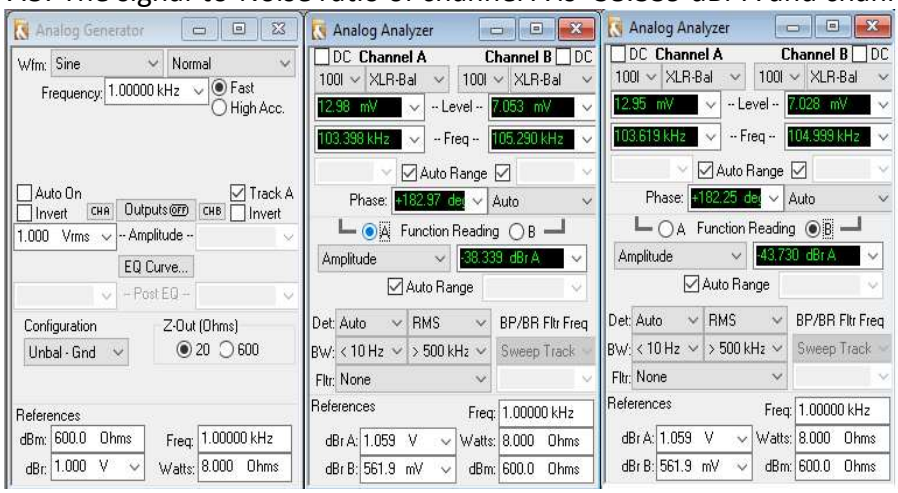


7.4. For 1kHz, crosstalk from B into A is +5.497 dB, and crosstalk from A into B is -5.506 dB. For 10kHz, crosstalk from B into A is +5.678 dB, and crosstalk from A into B is -5.752 dB. All crosstalk values are greater than -50 dB which means that channel separation is not very good.





7.5. The signal-to-Noise ratio of channel A is -38.339 dBr A and channel B is -43.730 dBr A.



8. Comparison between measurement results and the manufacturers' specifications.

	Specified on manual	Measured value
Frequency response	20 Hz to 20 kHz, ±0.1 dB	Mostly flat with some unevenness from 1k to 5k
THD+N	<0.003%	Channel A: 2.07423% Channel B: 1.92751%
Phase at 100Hz	Not Specified	Channel A: -0.14 deg Channel B: -0.08 deg
Phase at 1kHz	Not Specified	Channel A: -0.65 deg Channel B: -0.64 deg
Phase at 10kHz	Not Specified	Channel A: +0.12 deg Channel B: +0.13 deg
Crosstalk at 1kHz	Not specified	Channel A to B: -5.506 dB Channel B to A: +5.497 dB
Crosstalk at 10kHz	Not specified	Channel A to B: -5.752 dB Channel B to A: +5.678 dB
SNR	Not specified	Channel A: -38.339 dBr A Channel B: -43.730 dBr A

For the Focusrite Scarlett 2i2, the values specified on the user manual do not align with the values we measured. This may have been because of the age of this device being very old. The measured frequency response was mostly flat, but some unevenness could be seen in the mid-range. Measured values for THD+N for both channels A and B were well beyond the specification of 0.003% suggesting that overtime, the harmonic distortions may have increased. There were no specifications for phase, but all the phase values at 100Hz, 1kHz, and 10kHz were less than 1 degree which suggests that both channels are in phase. The crosstalk was also not specified in the manual, but we can see that there is crosstalk from both channel A to B and B to A. This amount of crosstalk is significant and shows the deterioration of the device. There is no specified SNR, and the measured SNR is -38.34 dBr for channel A and -43.73 dBr for channel B. In contrast to the previous device there is significantly less SNR.

Device Under Test 3: Behringer Xenyx 802

9. Manufacturer's Specifications

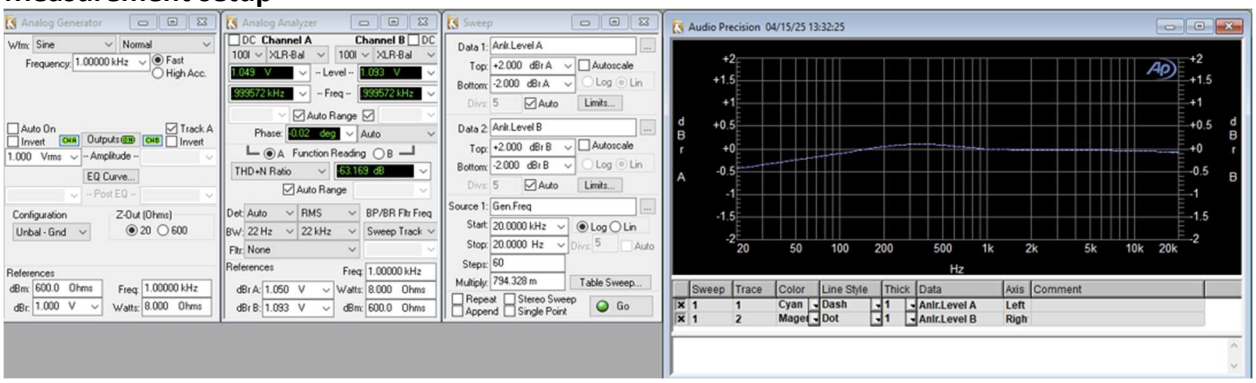
- Frequency response:
  <10 Hz – 150 kHz: -1 dB
  <10 Hz – 200 kHz: -3 dB
- THD+N: 0.005% / 0.004% A-weighted
- Phase: Not specified
- Crosstalk:
  Main fader closed: 90 dB

Channel muted: 89.5 dB
Channel fader muted: 89 dB

- Signal-to-Noise ratio: 110 dB / 112 dB A-weighted (0 dBu In @ +22 dB Gain)

Reference: https://mediadl.musictribe.com/media/sys_master/hcb/hea/8849649041438.pdf

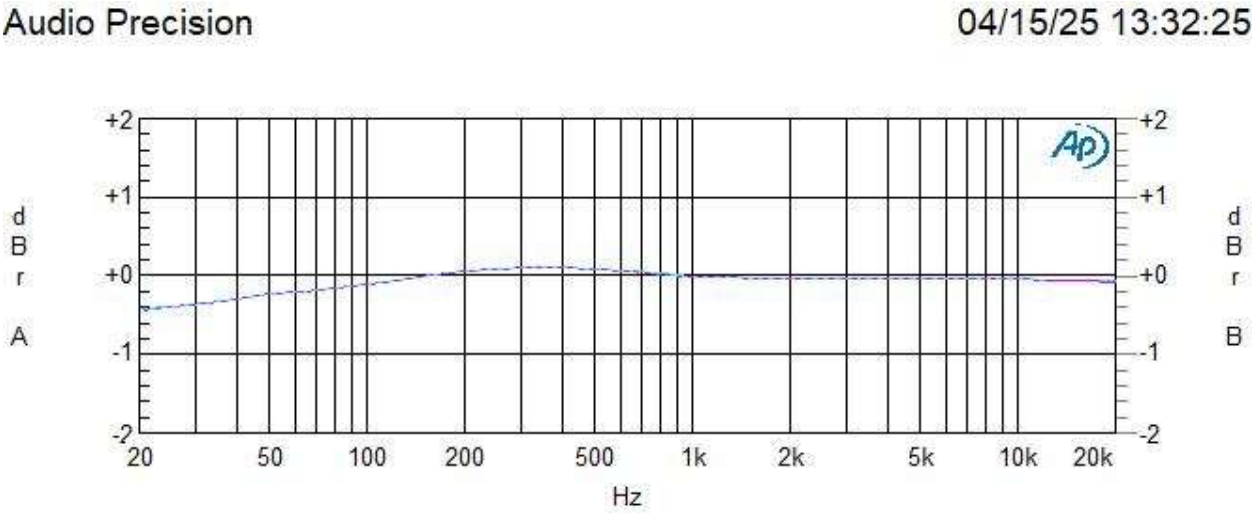
10. Measurement setup



After noticing that the analyzer settings for Scarlett 2i2 were not correct for the last few measurements, we reconvened to do another measurement. However, this device was not available, so we decided to measure the Behringer instead. Like before, we changed "configuration" on the analog generator to "Unbal – Gnd". Outputs were turned on for both channels. For the analyzer, we turned the pre-amp up until we reached close to 1V on levels for both channels. Changed bandwidth (BW) to 22Hz – 22kHz then went to edit tab in ribbon to "Set Analyzer dBr Ref" and "Set generator dBr Ref". For sweep panel, data 1 was set to "Anlr.LevelA", and data 2 was set to "Anlr.LevelB". Changed top and bottom for both data 1 and 2 to be dBr units. Changed steps to 60 then clicked on "Go".

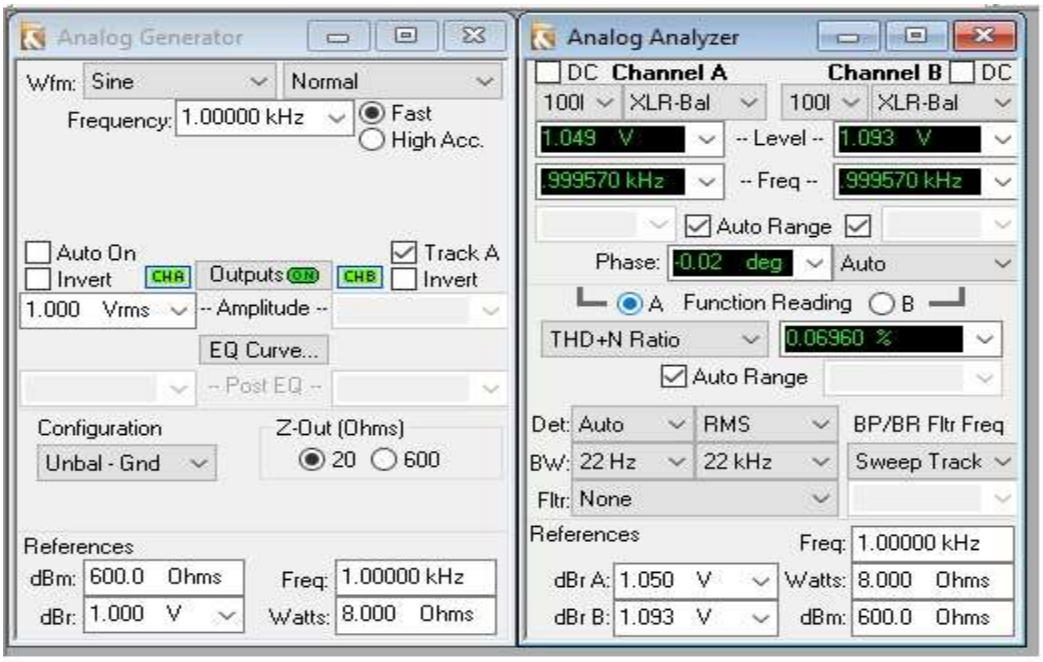
11. Results

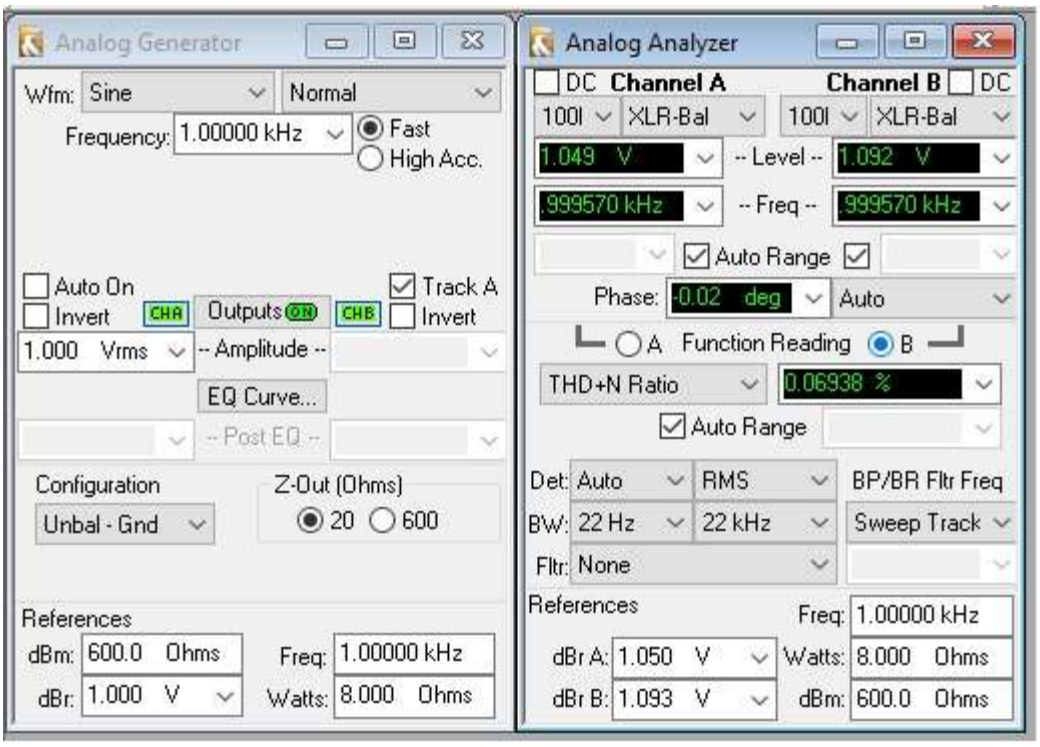
11.1. Frequency response of Ch 1 & Ch 2 shows a steady flat line, with low shelf below 150Hz. The level difference between the two channels was extremely tiny and we can see that the two lines (dash and dot) overlap for most parts on the graph.



Sweep	Trace	Color	Line Style	Thick	Data	Axis	Comment
1	1	Cyan	Dash	1	Anlr.Level A	Left	
1	2	Magenta	Dot	1	Anlr.Level B	Right	

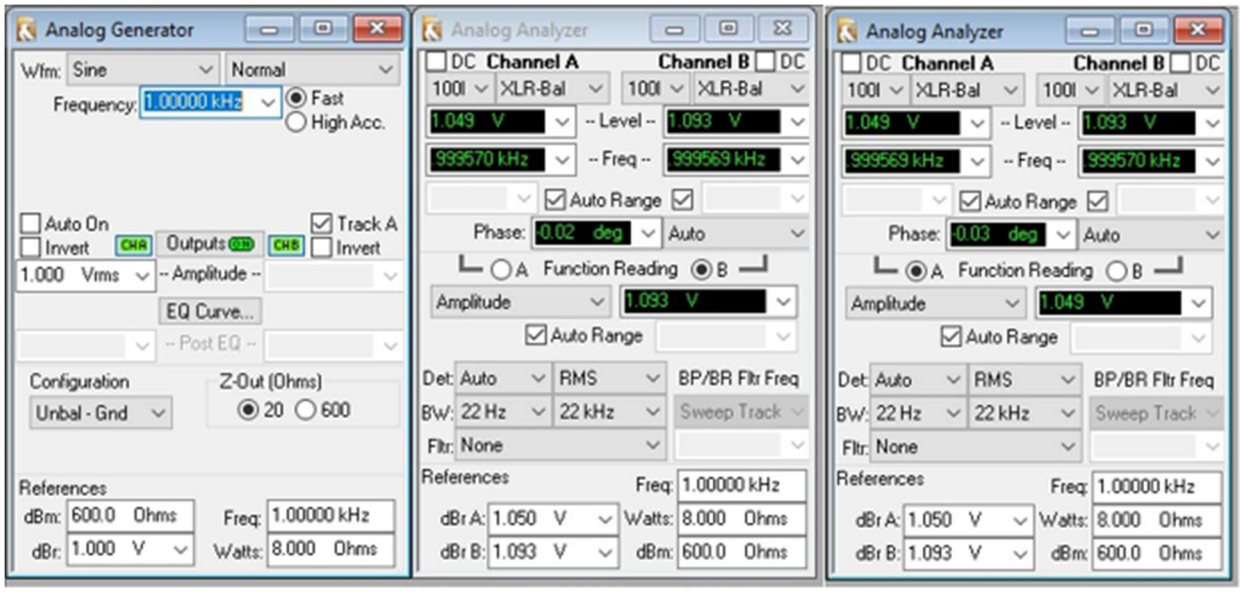
11.2. THD+N of channel A is 0.06960%, and channel B is 0.06938%.





11.3. The phase reading from both channel A and B at 100Hz is +0.01 deg. For 1kHz, phase reading of channel A is -0.02 deg and channel B is -0.03 deg. And for 10kHz, phase reading of both channel A and channel B is -0.20 deg. The two channels are in phase and the degree of separation is less than 1 degree.



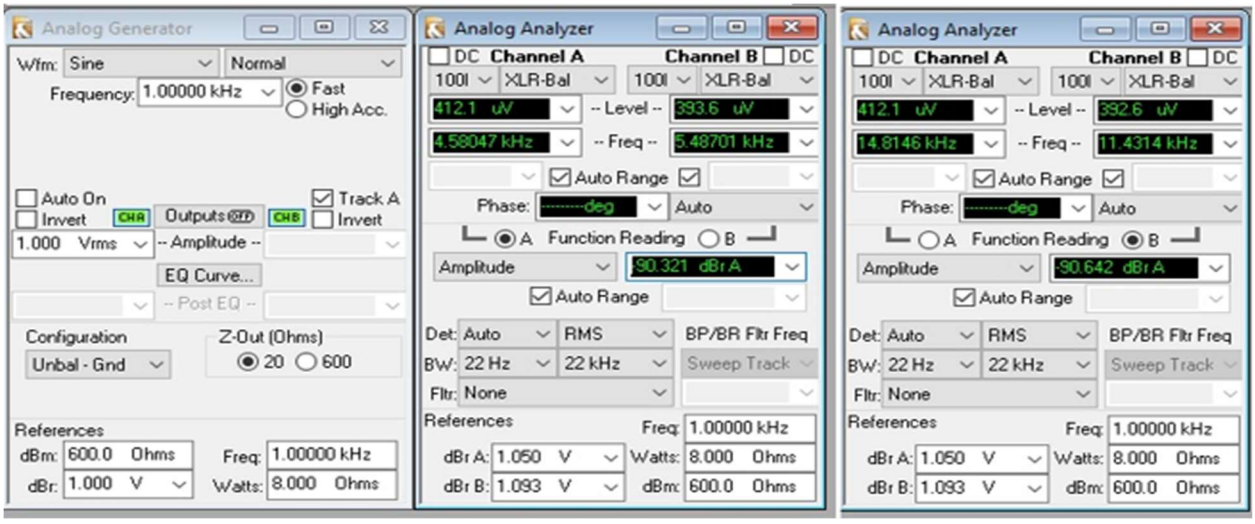

Analog Generator
Wfm: Sine
Normal
Frequency: 1.00000 kHz
Fast
High Acc.
Auto On
Invert
CHA
Outputs
CHB
Track A
Invert
1.000 Vrms
-- Amplitude --
EQ Curve...
-- Post EQ --
Configuration
Unbal - Gnd
Z-Out (Ohms)
20
600
References
dBm: 600.0 Ohms
Freq: 1.00000 kHz
dBr: 1.000 V
Watts: 8.000 Ohms
Analog Analyzer
DC Channel A
Channel B DC
100I XLR-Bal
1.049 V
-- Level --
1.093 V
999570 kHz
-- Freq --
999569 kHz
Auto Range
Phase: 0.02 deg
Auto
A Function Reading B
Amplitude
1.093 V
Auto Range
Det: Auto
RMS
BP/BR Fltr Freq
BW: 22 Hz
22 kHz
Sweep Track
Fltr: None
References
Freq: 1.00000 kHz
dBr A: 1.050 V
Watts: 8.000 Ohms
dBr B: 1.093 V
dBm: 600.0 Ohms
Analog Analyzer
DC Channel A
Channel B DC
100I XLR-Bal
1.049 V
-- Level --
1.093 V
999569 kHz
-- Freq --
999570 kHz
Auto Range
Phase: 0.03 deg
Auto
A Function Reading B
Amplitude
1.049 V
Auto Range
Det: Auto
RMS
BP/BR Fltr Freq
BW: 22 Hz
22 kHz
Sweep Track
Fltr: None
References
Freq: 1.00000 kHz
dBr A: 1.050 V
Watts: 8.000 Ohms
dBr B: 1.093 V
dBm: 600.0 Ohms


Analog Generator
Wfm: Sine
Normal
Frequency: 10.0000 kHz
Fast
High Acc.
Auto On
Invert
CHA
Outputs
CHB
Track A
Invert
1.000 Vrms
-- Amplitude --
EQ Curve...
-- Post EQ --
Configuration
Unbal - Gnd
Z-Out (Ohms)
20
600
References
dBm: 600.0 Ohms
Freq: 1.00000 kHz
dBr: 1.000 V
Watts: 8.000 Ohms
Analog Analyzer
DC Channel A
Channel B DC
100I XLR-Bal
1.044 V
-- Level --
1.087 V
9.99639 kHz
-- Freq --
9.99639 kHz
Auto Range
Phase: -0.20 deg
Auto
A Function Reading B
Amplitude
1.051 V
Auto Range
Det: Auto
RMS
BP/BR Fltr Freq
BW: 22 Hz
22 kHz
Sweep Track
Fltr: None
References
Freq: 1.00000 kHz
dBr A: 1.050 V
Watts: 8.000 Ohms
dBr B: 1.093 V
dBm: 600.0 Ohms
Analog Analyzer
DC Channel A
Channel B DC
100I XLR-Bal
1.044 V
-- Level --
1.087 V
9.99639 kHz
-- Freq --
9.99639 kHz
Auto Range
Phase: -0.20 deg
Auto
A Function Reading B
Amplitude
1.095 V
Auto Range
Det: Auto
RMS
BP/BR Fltr Freq
BW: 22 Hz
22 kHz
Sweep Track
Fltr: None
References
Freq: 1.00000 kHz
dBr A: 1.050 V
Watts: 8.000 Ohms
dBr B: 1.093 V
dBm: 600.0 Ohms

11.4. For 1kHz, crosstalk from B into A is -0.375 dB, and crosstalk from A into B is -8.635 dB. For 10kHz, crosstalk from B into A is -0.341 dB, and crosstalk from A into B is -9.584 dB. More crosstalk can be seen from A to B than compared to B to A.



11.5. The signal-to-Noise ratio of channels A is -90.321 dBr A and channel B is -90.642 dBr A.



12. Comparison between your measurement results and the manufacturers' specifications. Briefly explain what differences (or similarities) are found.

	Specified on manual	Measured value
Frequency response	<10 Hz – 150 kHz: -1 dB <10 Hz – 200 kHz: -3 dB	Mostly flat, with low shelf below 150Hz
THD+N	0.005% / 0.004%	Channel A: 0.06960% Channel B: 0.06938%
Phase at 100Hz	Not Specified	Channel A: +0.01 deg Channel B: +0.01 deg
Phase at 1kHz	Not Specified	Channel A: -0.02 deg Channel B: -0.03 deg
Phase at 10kHz	Not Specified	Channel A: -0.03 deg Channel B: -0.03 deg
Crosstalk at 1kHz	89-90 dB	Channel A to B: -8.635 dB Channel B to A: -0.375 dB
Crosstalk at 10kHz	89-90 dB	Channel A to B: -9.584 dB Channel B to A: -0.341 dB
SNR	110 dB / 112 dB	Channel A: -90.321 dBr A Channel B: -90.642 dBr A

For the Behringer Xenyx 802, the values specified on the user manual mostly do not align with the values we measured. The measured frequency response shows a steady flat line with a 0.2 dB low shelf. Measured values for THD+N for both channels A and B are slightly above the specification which suggests that the device has slightly deteriorated because of age. There were no specifications for phase, but all the phase values at 100Hz, 1kHz, and 10kHz are extremely tiny which suggests that both channels are in phase. The specified crosstalk was 89-90 dB, but the measured values are greater than this specification. Moreover, there is more crosstalk from A to B which suggests that channel B may have more damage than channel A. The specified SNR was 110-112 dB, but the measured values are roughly 90 dB. This suggests that system noise may have increased or the passing of signal strength may have decreased.